Assignment 2: Data Structures and Methods for HPC

Group 15

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Imports

```
import ipytest
import pytest
import random
import time
import numpy as np
import scipy.stats
from IPython.display import display, Markdown


ipytest.autoconfig()
```

Julia Set Code

```
# area of complex space to investigate
x1, x2, y1, y2 = -1.8, 1.8, -1.8, 1.8
c_real, c_imag = -0.62772, -0.42193

def calc_pure_python(desired_width, max_iterations):
    """Create a list of complex coordinates (zs) and complex parameters (cs),
    build Julia set"""
    x_step = (x2 - x1) / desired_width
    y_step = (y1 - y2) / desired_width
    x = []
    y = []
    ycoord = y2
    while ycoord > y1:
        y.append(ycoord)
        ycoord += y_step
    xcoord = x1
    while xcoord < x2:
        x.append(xcoord)
        xcoord += x_step
    # build a list of coordinates and the initial condition for each cell.
    # Note that our initial condition is a constant and could easily be removed,
    # we use it to simulate a real-world scenario with several inputs to our
    # function
    zs = []
    cs = []
    for ycoord in y:
        for xcoord in x:
            zs.append(complex(xcoord, ycoord))
            cs.append(complex(c_real, c_imag))

    print("Length of x:", len(x))
    print("Total elements:", len(zs))
    start_time = time.time()
    output = calculate_z_serial_purepython(max_iterations, zs, cs)
    end_time = time.time()
    secs = end_time - start_time
    print(calculate_z_serial_purepython.__name__ + " took", secs, "seconds")

    # This sum is expected for a 1000^2 grid with 300 iterations
    # It ensures that our code evolves exactly as we'd intended
    # assert sum(output) == 33219980
    return output

def calculate_z_serial_purepython(maxiter, zs, cs):
    """Calculate output list using Julia update rule"""
    output = [0] * len(zs)
```

```
    for i in range(len(zs)):
        n = 0
        z = zs[i]
        c = cs[i]
        while abs(z) < 2 and n < maxiter:
            z = z * z + c
            n += 1
        output[i] = n
    return output

if __name__ == "__main__":
    # Calculate the Julia set using a pure Python solution with
    # reasonable defaults for a laptop
    calc_pure_python(desired_width=1000, max_iterations=300)
```

```
calculate_z_serial_purepython took 1.8392739295959473 seconds
```

Exercise 1: PyTest with the Julia Set Code

Task 1.1: Implement a separate code to test the assertion above using the pytest framework

An extra return statement has been added to the `calc_pure_python` function to return the output list. The test function `test_output_sum_for_300_iterations` uses pytest to assert that the sum of the output list for 300 iterations is equal to 33219980.

> *The test function below uses the `ipytest` magic command to run pytest within a Jupyter notebook cell. In normal Python files, a `@pytest` decorator would be used instead.*

```
%%ipytest -qq

def test_output_sum_for_300_iterations():
    assert sum(
        calc_pure_python(desired_width=1000, max_iterations=300)
    ) == 33219980
```

Task 1.2: How would you implement the unit test with the possibility of having a different number of iterations and grid points?

The above test function in Task 1.1 can be modified to be parameterised as follows:

```
@pytest.mark.parametrize(
    "width, iterations, expected_sum",
    [
        (1000, 300, 33219980)
    ],
)
def test_output_sum_for_300_iterations_parameterised(width, iterations, expected_sum):
    assert sum(
        calc_pure_python(desired_width=width, max_iterations=iterations)
    ) == expected_sum
```

Exercise 2: Python DGEMM Benchmark Operation

Task 2.1: Implement the DGEMM with matrices as NumPy array

```
def dgemm_numpy(a: np.ndarray, b: np.ndarray, c: np.ndarray) -> None:
    m, p = a.shape
    n = b.shape[1]

    for i in range(m):
        for j in range(n):
            for k in range(p):
                c[i, j] += a[i, k] * b[k, j]
```

Task 2.2: Using pytest develop a unit test for checking the correctness

```
%%ipytest -qq
```

```python
def test_dgemm_numpy():
    a = np.array([[11.5, 3.007], [7.0, 11.0]])
    b = np.array([[8.0, 0.0, 1.0], [0.329, 3.0, 5.0]])
    c = np.array([[2.0, 3.0, 9.0], [7.912, 1.0, 2.0]])
    expected_c = np.array([[94.989303, 12.021, 35.535], [67.531, 34.0, 64.0]])
    dgemm_numpy(a, b, c)
    np.testing.assert_array_almost_equal(c, expected_c)
```

Task 2.3: Measure the execution time for each approach varying the matrix size

Code for implementing with Python lists

```python
def dgemm_list(a: list, b: list, c: list) -> None:
    m = len(a)
    p = len(a[0])
    n = len(b[0])

    for i in range(m):
        for j in range(n):
            for k in range(p):
                c[i][j] += a[i][k] * b[k][j]
```

```python
%%ipytest -qq

def test_dgemm_list():
    a = [[11.5, 3.007], [7.0, 11.0]]
    b = [[8.0, 0.0, 1.0], [0.329, 3.0, 5.0]]
    c = [[2.0, 3.0, 9.0], [7.912, 1.0, 2.0]]
    expected_c = [[94.989303, 12.021, 35.535], [67.531, 34.0, 64.0]]
    dgemm_list(a, b, c)
    np.testing.assert_array_almost_equal(c, expected_c)
```

Code for implementing with Python tuples (arrays)

```python
def dgemm_tuple(a: tuple, b: tuple, c: tuple) -> tuple:
    m = len(a)
    p = len(a[0])
    n = len(b[0])

    return tuple(
        tuple(
            c[i][j] + sum(a[i][k] * b[k][j] for k in range(p))
            for j in range(n)
        )
        for i in range(m)
    )
```

```python
%%ipytest -qq

def test_dgemm_tuple():
    a = ((11.5, 3.007), (7.0, 11.0))
    b = ((8.0, 0.0, 1.0), (0.329, 3.0, 5.0))
    c = ((2.0, 3.0, 9.0), (7.912, 1.0, 2.0))
    expected_c = ((94.989303, 12.021, 35.535), (67.531, 34.0, 64.0))
    new_c = dgemm_tuple(a, b, c)
    np.testing.assert_array_almost_equal(new_c, expected_c)
```

Benchmarking

> *The following benchmark function helps iterate dgemm with different implementations.*

```python
def benchmark_sweep(dgemm_func, init_func, name, start=5, end=100, step=5, iterations=20):

    sizes = list(range(start, end + 1, step))
    means, mins, maxs = [], [], []
    cis = [] # Margin of error for CI
    stds = []
```

```
    table_lines = [
        f"### Detailed Results: {name}",
        "| Matrix Size | Average (s) | Std Dev (s) | Min (s) | Max (s) | 95% CI Margin (s) |",
        "|:---:|:---:|:---:|:---:|:---:|:---:|"
    ]

    print(f"Running benchmark sweep for {name}...")

    for n in sizes:
        durations = []
        for _ in range(iterations):
            # Initialize matrices using the provided helper
            a, b, c = init_func(n)

            start_time = time.time()
            dgemm_func(a, b, c)
            end_time = time.time()

            durations.append(end_time - start_time)

        # Statistics
        arr = np.array(durations)
        avg = np.mean(arr)
        std = np.std(arr, ddof=1)
        means.append(avg)
        mins.append(np.min(arr))
        maxs.append(np.max(arr))
        stds.append(std)

        # 95% Confidence Interval calculation
        # CI = mean +/- margin_of_error
        # margin_of_error = t * (std / sqrt(n))
        dof = iterations - 1
        t_crit = np.abs(scipy.stats.t.ppf((1 - 0.95) / 2, dof))
        margin_of_error = t_crit * (std / np.sqrt(iterations))
        cis.append(margin_of_error)

        row = f"| {n}x{n} | {avg:.4e} | {std:.4e} | {np.min(arr):.4e} | {np.max(arr):.4e} | ±{margin_of_error:.4e} |"
        table_lines.append(row)

    display(Markdown("\n".join(table_lines)))
```

Benchmarking for Python lists

Initialiser

```
def init_list_matrices(n):
    a = [[random.random() for _ in range(n)] for _ in range(n)]
    b = [[random.random() for _ in range(n)] for _ in range(n)]
    c = [[0.0 for _ in range(n)] for _ in range(n)]
    return a, b, c
```

Benchmark Execution

```
benchmark_sweep(dgemm_list, init_list_matrices, "Python Lists", start=10, end=100, step=10, iterations=50)
```

Detailed Results: Python Lists

Matrix Size	Average (s)	Std Dev (s)	Min (s)	Max (s)	95% CI Margin (s)
10x10	8.1687e-05	9.9661e-06	7.1049e-05	1.2088e-04	±2.8323e-06
20x20	4.8196e-04	6.7472e-05	3.9196e-04	6.8784e-04	±1.9175e-05
30x30	1.1182e-03	9.5056e-05	9.7013e-04	1.3838e-03	±2.7015e-05
40x40	2.4643e-03	9.3585e-05	2.2547e-03	2.7499e-03	±2.6596e-05
50x50	4.9397e-03	1.8874e-04	4.5300e-03	5.4290e-03	±5.3638e-05
60x60	8.3588e-03	1.2205e-04	8.0299e-03	8.7471e-03	±3.4685e-05
70x70	1.3137e-02	2.1648e-04	1.2641e-02	1.3634e-02	±6.1523e-05
80x80	1.9637e-02	2.5855e-04	1.8815e-02	2.0101e-02	±7.3479e-05

Matrix Size	Average (s)	Std Dev (s)	Min (s)	Max (s)	95% CI Margin (s)
90x90	2.8377e-02	6.5097e-04	2.7314e-02	2.9595e-02	±1.8500e-04
100x100	3.8517e-02	5.4605e-04	3.7727e-02	4.0397e-02	±1.5519e-04

Benchmarking for Python tuples

Initialiser

```
def init_tuple_matrices(n):
    """Initializes matrices as tuples for the requested size n."""
    a = tuple(tuple(random.random() for _ in range(n)) for _ in range(n))
    b = tuple(tuple(random.random() for _ in range(n)) for _ in range(n))
    c = tuple(tuple(0.0 for _ in range(n)) for _ in range(n))
    return a, b, c
```

Benchmark Execution

```
benchmark_sweep(dgemm_tuple, init_tuple_matrices, "Python Tuples", start=10, end=100, step=10, iterations=50)
```

Detailed Results: Python Tuples

Matrix Size	Average (s)	Std Dev (s)	Min (s)	Max (s)	95% CI Margin (s)
10x10	1.0435e-04	2.6962e-05	9.1791e-05	2.4700e-04	±7.6626e-06
20x20	4.8028e-04	6.3528e-05	3.9411e-04	6.8784e-04	±1.8055e-05
30x30	1.0978e-03	7.6868e-05	9.7513e-04	1.2569e-03	±2.1846e-05
40x40	2.4012e-03	8.4583e-05	2.2280e-03	2.7461e-03	±2.4038e-05
50x50	4.6327e-03	1.3172e-04	4.3452e-03	4.8969e-03	±3.7436e-05
60x60	7.9803e-03	2.2889e-04	7.5471e-03	8.4639e-03	±6.5048e-05
70x70	1.2452e-02	3.2135e-04	1.2045e-02	1.3240e-02	±9.1326e-05
80x80	1.8023e-02	2.8103e-04	1.7452e-02	1.8982e-02	±7.9867e-05
90x90	2.5635e-02	5.2794e-04	2.5105e-02	2.7178e-02	±1.5004e-04
100x100	3.4453e-02	1.7509e-04	3.4091e-02	3.5347e-02	±4.9760e-05

Benchmarking for NumPy arrays

Initialiser

```
def init_numpy_matrices(n):
    """Initializes matrices as NumPy arrays for the requested size n."""
    a = np.random.rand(n, n)
    b = np.random.rand(n, n)
    c = np.zeros((n, n))
    return a, b, c
```

Benchmark Execution

```
benchmark_sweep(dgemm_numpy, init_numpy_matrices, "NumPy Arrays", start=10, end=100, step=10, iterations=50)
```

Detailed Results: NumPy Arrays

Matrix Size	Average (s)	Std Dev (s)	Min (s)	Max (s)	95% CI Margin (s)
10x10	2.9938e-04	3.4639e-05	2.5010e-04	3.8910e-04	±9.8442e-06
20x20	1.6458e-03	1.5185e-04	1.5182e-03	2.1276e-03	±4.3156e-05
30x30	5.2404e-03	1.4341e-04	5.0781e-03	5.7080e-03	±4.0755e-05
40x40	1.2726e-02	3.7605e-04	1.2066e-02	1.4115e-02	±1.0687e-04
50x50	2.4407e-02	6.7536e-04	2.3508e-02	2.5853e-02	±1.9194e-04
60x60	4.3653e-02	5.2564e-03	4.0631e-02	6.2964e-02	±1.4939e-03
70x70	7.9405e-02	2.8232e-02	6.4548e-02	2.0552e-01	±8.0235e-03
80x80	9.7417e-02	6.7181e-04	9.5891e-02	9.9707e-02	±1.9093e-04

Matrix Size	Average (s)	Std Dev (s)	Min (s)	Max (s)	95% CI Margin (s)
90x90	1.3842e-01	9.3904e-04	1.3693e-01	1.4129e-01	±2.6687e-04
100x100	1.9191e-01	2.5164e-03	1.8833e-01	2.0141e-01	±7.1514e-04

Benchmark Results Analysis

The results indicate that Python lists and tuples yield comparable average computation times. Contrary to common expectation, NumPy arrays do not outperform either structure, despite their optimised C implementation. This arises because the nested loops in the DGEMM implementation bypass NumPy's vectorised operations, resulting in performance akin to that of native Python data structures. Indeed, direct element access in NumPy arrays proves slower than in lists or tuples, with computation times approximately one order of magnitude higher. This might be attributed to the overhead of NumPy's internal handling of array accesses, having to create temporary Python `float` objects for each element access.

As matrix dimensions increase, both the mean computation time and the standard deviation rise. The growth in mean time is expected, given the $O(n^3)$ complexity of the algorithm. The increase in standard deviation and consequently the widening of the confidence interval likely comes from greater susceptibility to external interference during longer executions. Such interference may include additional CPU context switches, scheduling delays, or other system-level activity. For larger matrices, cache misses and increased main-memory access might further contribute to execution-time variability.

Task 2.4: Using the timing information and the number of operations for the DGEMM, calculate the FLOPS/s

In each of the DGEMM implementations, the total number of floating-point operations (FLOPs) for multiplying two n x n matrices is given by $2n^3$ as each loop iteration contains one floating-point multiplication and one floating-point addition.

Taking the average execution time of a 100 x 100 matrix multiplication using Python Lists as an example, which is approximately 0.038517 seconds, we can calculate the FLOPS/s as follows:

$$\text{FLOPS/s} = \frac{2(100)^3}{0.038517} \approx 51925124 \text{ FLOPS/s}$$

All the results for this task are done on Apple M4 chip. Assuming that one operation is done per cycle, then the theoretical peak is the clock frequency value: 4.3 GHz or 4.3 x 10^9 FLOPS/s.

This implies that the achieved performance of approximately 51.9 MFLOPS/s is about 1.2% of the theoretical peak performance of the Apple M4 chip and one floating point operation might not be done per cycle.

Task 2.5: Compare the performance results with the numpy matmul operation

```
def dgemm_numpy_matmul(a: np.ndarray, b: np.ndarray, c: np.ndarray) -> None:
    c += np.matmul(a, b)
```

Benchmark Execution

```
benchmark_sweep(dgemm_numpy_matmul, init_numpy_matrices, "NumPy Arrays with matmul", start=10, end=100, step=10,
iterations=50)
```

Detailed Results: NumPy Arrays with matmul

Matrix Size	Average (s)	Std Dev (s)	Min (s)	Max (s)	95% CI Margin (s)
10x10	2.9278e-06	6.3969e-06	7.1526e-07	3.2187e-05	±1.8180e-06
20x20	3.3760e-06	6.5312e-06	9.5367e-07	4.2915e-05	±1.8561e-06
30x30	2.7084e-06	3.4933e-06	9.5367e-07	2.6226e-05	±9.9279e-07
40x40	3.1185e-06	2.5442e-06	1.6689e-06	1.5974e-05	±7.2305e-07
50x50	4.7922e-06	5.0655e-06	2.8610e-06	3.1948e-05	±1.4396e-06
60x60	4.2772e-06	6.0433e-07	3.8147e-06	7.1526e-06	±1.7175e-07
70x70	6.1131e-06	7.7183e-07	4.7684e-06	9.0599e-06	±2.1935e-07
80x80	6.8426e-06	8.1352e-07	5.7220e-06	1.0967e-05	±2.3120e-07
90x90	9.5367e-06	2.0345e-06	8.8215e-06	2.3127e-05	±5.7819e-07
100x100	1.1983e-05	1.3425e-06	1.0967e-05	1.8120e-05	±3.8153e-07

Benchmark Results Analysis for NumPy matmul

The NumPy `matmul` function significantly outperforms the custom DGEMM implementation using NumPy arrays, tuples and lists. This is expected, as `matmul` leverages optimised low-level libraries (BLAS) that are vectorised. The performance difference becomes more evident with increasing matrix sizes, despite the runtime for smaller matrices remain similar. It can also be observed that the standard deviation and confidence intervals for `matmul` remains relatively constant across matrix sizes, indicating consistent performance likely due to efficient memory access patterns and optimised algorithms.

Exercise 3: Experiment with the Python Debugger

Task 3.1: What are the advantages of using a debugger? What challenges did you find in using the pdb debugger, if any?

One of the advantages of using a debugger is that one can inspect the state of variables at any point during execution, eliminating the need for programmers to rely on print statements. Additionally, it enables programmers to step through the code line by line, placing the flow of execution under a direct control.

Although this is one of my first experiences using the Python debugger, I found it very easy to use, given my prior experience with LLDB and GDB. It also provides useful pretty-printing functionality. However, compared to LLDB, the debugger appears less feature-rich, lacking capabilities such as reverse stepping. Overall, I encountered no significant challenges when using pdb.

Bonus Exercise: Performance Analysis and Optimization of the Game of Life Code

Task B.1

Pylint

```
pylint conway.py
************* Module conway
conway.py:11:31: C0303: Trailing whitespace (trailing-whitespace)
conway.py:24:38: C0303: Trailing whitespace (trailing-whitespace)
conway.py:25:38: C0303: Trailing whitespace (trailing-whitespace)
conway.py:59:28: C0303: Trailing whitespace (trailing-whitespace)
conway.py:64:70: C0303: Trailing whitespace (trailing-whitespace)
conway.py:66:62: C0303: Trailing whitespace (trailing-whitespace)
conway.py:67:62: C0303: Trailing whitespace (trailing-whitespace)
conway.py:68:74: C0303: Trailing whitespace (trailing-whitespace)
conway.py:95:0: C0303: Trailing whitespace (trailing-whitespace)
conway.py:100:0: C0303: Trailing whitespace (trailing-whitespace)
conway.py:127:18: C0303: Trailing whitespace (trailing-whitespace)
conway.py:80:0: R1707: Disallow trailing comma tuple (trailing-comma-tuple)
conway.py:9:0: C0410: Multiple imports on one line (sys, argparse) (multiple-imports)
conway.py:12:0: R0402: Use 'from matplotlib import animation' instead (consider-using-from-import)
conway.py:18:0: C0103: Function name "randomGrid" doesn't conform to snake_case naming style (invalid-name)
conway.py:18:15: C0103: Argument name "N" doesn't conform to snake_case naming style (invalid-name)
conway.py:22:0: C0103: Function name "addGlider" doesn't conform to snake_case naming style (invalid-name)
conway.py:29:0: C0103: Function name "addGosperGliderGun" doesn't conform to snake_case naming style (invalid-name)
conway.py:57:0: C0116: Missing function or method docstring (missing-function-docstring)
conway.py:57:11: C0103: Argument name "frameNum" doesn't conform to snake_case naming style (invalid-name)
conway.py:57:32: C0103: Argument name "N" doesn't conform to snake_case naming style (invalid-name)
conway.py:60:4: C0103: Variable name "newGrid" doesn't conform to snake_case naming style (invalid-name)
conway.py:57:11: W0613: Unused argument 'frameNum' (unused-argument)
conway.py:83:0: C0116: Missing function or method docstring (missing-function-docstring)
conway.py:97:4: C0103: Variable name "N" doesn't conform to snake_case naming style (invalid-name)
conway.py:99:8: C0103: Variable name "N" doesn't conform to snake_case naming style (invalid-name)
conway.py:102:4: C0103: Variable name "updateInterval" doesn't conform to snake_case naming style (invalid-name)
conway.py:104:8: C0103: Variable name "updateInterval" doesn't conform to snake_case naming style (invalid-name)
conway.py:9:0: W0611: Unused import sys (unused-import)

------------------------------------
Your code has been rated at 6.18/10
```

Black

```
"""
conway_original.py

A simple Python/matplotlib implementation of Conway's Game of Life.

Author: Mahesh Venkitachalam
"""
```

```python
import sys, argparse
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation

ON = 255
OFF = 0
vals = [ON, OFF]


def randomGrid(N):
    """returns a grid of NxN random values"""
    return np.random.choice(vals, N * N, p=[0.2, 0.8]).reshape(N, N)


def addGlider(i, j, grid):
    """adds a glider with top left cell at (i, j)"""
    glider = np.array([[0, 0, 255], [255, 0, 255], [0, 255, 255]])
    grid[i : i + 3, j : j + 3] = glider


def addGosperGliderGun(i, j, grid):
    """adds a Gosper Glider Gun with top left cell at (i, j)"""
    gun = np.zeros(11 * 38).reshape(11, 38)

    gun[5][1] = gun[5][2] = 255
    gun[6][1] = gun[6][2] = 255

    gun[3][13] = gun[3][14] = 255
    gun[4][12] = gun[4][16] = 255
    gun[5][11] = gun[5][17] = 255
    gun[6][11] = gun[6][15] = gun[6][17] = gun[6][18] = 255
    gun[7][11] = gun[7][17] = 255
    gun[8][12] = gun[8][16] = 255
    gun[9][13] = gun[9][14] = 255

    gun[1][25] = 255
    gun[2][23] = gun[2][25] = 255
    gun[3][21] = gun[3][22] = 255
    gun[4][21] = gun[4][22] = 255
    gun[5][21] = gun[5][22] = 255
    gun[6][23] = gun[6][25] = 255
    gun[7][25] = 255

    gun[3][35] = gun[3][36] = 255
    gun[4][35] = gun[4][36] = 255

    grid[i : i + 11, j : j + 38] = gun

def step(grid, N):
    #for profiling only
    newGrid = grid.copy()
    for i in range(N):
        for j in range(N):
            total = int(
                (
                    grid[i, (j - 1) % N]
                    + grid[i, (j + 1) % N]
                    + grid[(i - 1) % N, j]
                    + grid[(i + 1) % N, j]
                    + grid[(i - 1) % N, (j - 1) % N]
                    + grid[(i - 1) % N, (j + 1) % N]
                    + grid[(i + 1) % N, (j - 1) % N]
                    + grid[(i + 1) % N, (j + 1) % N]
                )
                / 255
            )
            if grid[i, j] == ON:
                if (total < 2) or (total > 3):
                    newGrid[i, j] = OFF
            else:
```

```
                if total == 3:
                    newGrid[i, j] = ON
    return newGrid

def update(frameNum, img, grid, N):
    # copy grid since we require 8 neighbors for calculation
    # and we go line by line
    newGrid = grid.copy()
    for i in range(N):
        for j in range(N):
            # compute 8-neghbor sum
            # using toroidal boundary conditions - x and y wrap around
            # so that the simulaton takes place on a toroidal surface.
            total = int(
                (
                    grid[i, (j - 1) % N]
                    + grid[i, (j + 1) % N]
                    + grid[(i - 1) % N, j]
                    + grid[(i + 1) % N, j]
                    + grid[(i - 1) % N, (j - 1) % N]
                    + grid[(i - 1) % N, (j + 1) % N]
                    + grid[(i + 1) % N, (j - 1) % N]
                    + grid[(i + 1) % N, (j + 1) % N]
                )
                / 255
            )
            # apply Conway's rules
            if grid[i, j] == ON:
                if (total < 2) or (total > 3):
                    newGrid[i, j] = OFF
            else:
                if total == 3:
                    newGrid[i, j] = ON
    # update data
    img.set_data(newGrid)
    grid[:] = newGrid[:]
    return (img,)


# main() function
def main():
    # Command line args are in sys.argv[1], sys.argv[2] ..
    # sys.argv[0] is the script name itself and can be ignored
    # parse arguments
    parser = argparse.ArgumentParser(
        description="Runs Conway's Game of Life simulation."
    )
    # add arguments
    parser.add_argument("--grid-size", dest="N", required=False)
    parser.add_argument("--mov-file", dest="movfile", required=False)
    parser.add_argument("--interval", dest="interval", required=False)
    parser.add_argument("--glider", action="store_true", required=False)
    parser.add_argument("--gosper", action="store_true", required=False)
    args = parser.parse_args()

    # set grid size
    N = 100
    if args.N and int(args.N) > 8:
        N = int(args.N)

    # set animation update interval
    updateInterval = 50
    if args.interval:
        updateInterval = int(args.interval)

    # declare grid
    grid = np.array([])
    # check if "glider" demo flag is specified
    if args.glider:
        grid = np.zeros(N * N).reshape(N, N)
        addGlider(1, 1, grid)
    elif args.gosper:
```

```
        grid = np.zeros(N * N).reshape(N, N)
        addGosperGliderGun(10, 10, grid)
    else:
        # populate grid with random on/off - more off than on
        grid = randomGrid(N)

    # set up animation
    fig, ax = plt.subplots()
    img = ax.imshow(grid, interpolation="nearest")
    ani = animation.FuncAnimation(
        fig,
        update,
        fargs=(
            img,
            grid,
            N,
        ),
        frames=10,
        interval=updateInterval,
        save_count=50,
    )

    # # of frames?
    # set output file
    if args.movfile:
        ani.save(args.movfile, fps=30, extra_args=["-vcodec", "libx264"])

    plt.show()


# call main
if __name__ == "__main__":
    main()
```

Sphinx (HTML documentation)

```
sphinx-build -b html docs docs/_build/html
Running Sphinx v9.0.4
loading translations [en]... done
loading pickled environment... The configuration has changed (1 option: 'html_static_path')
done
building [mo]: targets for 0 po files that are out of date
writing output...
building [html]: build_info mismatch, copying .buildinfo to .buildinfo.bak
building [html]: targets for 1 source files that are out of date
updating environment: 0 added, 0 changed, 0 removed
reading sources...
looking for now-outdated files... none found
preparing documents... done
copying assets...
copying static files...
Writing evaluated template result to /Users/klaus/Documents/KTH/DD2358/dd2358-work/docs/_build/html/_static/basic.css
Writing evaluated template result to /Users/klaus/Documents/KTH/DD2358/dd2358-work/docs/_build/html/_static/language_data.js
Writing evaluated template result to /Users/klaus/Documents/KTH/DD2358/dd2358-
work/docs/_build/html/_static/documentation_options.js
Writing evaluated template result to /Users/klaus/Documents/KTH/DD2358/dd2358-work/docs/_build/html/_static/alabaster.css
copying static files: done
copying extra files...
copying extra files: done
copying assets: done
writing output... [100%] index
generating indices... genindex py-modindex done
writing additional pages... search done
dumping search index in English (code: en)... done
dumping object inventory... done
build succeeded.

The HTML pages are in docs/_build/html.
```

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- addGosperGliderGun() (in module conway)

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  - module

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Conway's Game of Life

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Conway’s Game of Life¶

A simple Python/matplotlib implementation of Conway’s Game of Life.

conway.py

A simple Python/matplotlib implementation of Conway’s Game of Life.

Author: Mahesh Venkitachalam

conway.addGlider(*i*, *j*, *grid*)¶

adds a glider with top left cell at (i, j)

conway.addGosperGliderGun(*i*, *j*, *grid*)¶

adds a Gosper Glider Gun with top left cell at (i, j)

conway.main()¶

conway.randomGrid(*N*)¶

returns a grid of NxN random values

conway.update(*frameNum*, *img*, *grid*, *N*)¶

Conway's Game of Life

Navigation

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- Documentation overview

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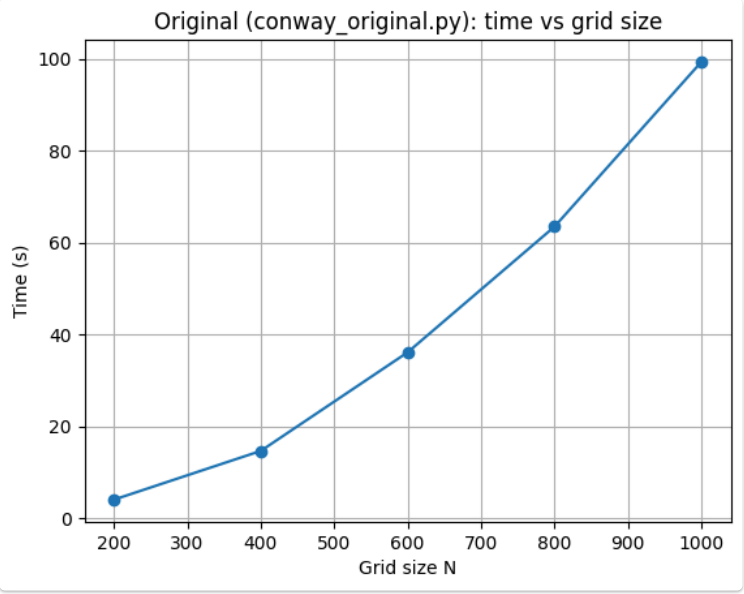
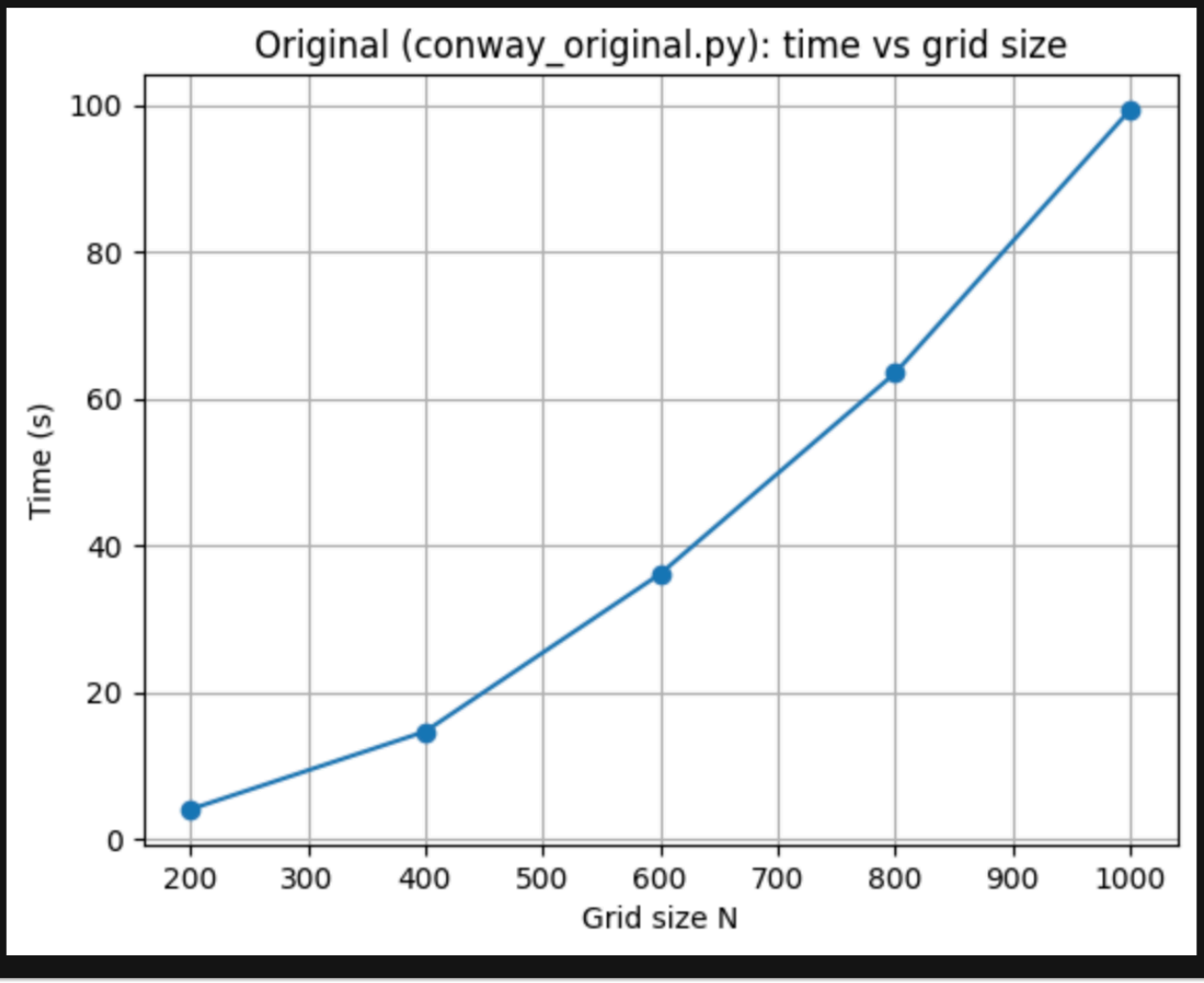
Task B.2

```python
import sys, os, time
import matplotlib.pyplot as plt

if os.path.isfile(os.path.join(os.getcwd(), "asm2", "conway_original.py")):
    sys.path.insert(0, os.path.join(os.getcwd(), "asm2"))
else:
    sys.path.insert(0, os.getcwd())
import conway_original

num_iterations = 100
grid_sizes = [200, 400, 600, 800, 1000]
times_sec = []
for N in grid_sizes:
    grid = conway_original.randomGrid(N)
    t0 = time.perf_counter()
    for _ in range(num_iterations):
        grid = conway_original.step(grid, N)
    t = time.perf_counter() - t0
    times_sec.append(t)
    print(f"N={N}: {t:.4f} s")
plt.plot(grid_sizes, times_sec, "o-")
plt.xlabel("Grid size N")
plt.ylabel("Time (s)")
plt.title("Original (conway_original.py): time vs grid size")
plt.grid(True)
plt.show()
```

```
N=200: 4.0130 s
N=400: 14.6322 s
N=600: 36.1137 s
N=800: 63.4908 s
N=1000: 99.3897 s
```



```
N=200: 4.0130 s
N=400: 14.6322 s
N=600: 36.1137 s
N=800: 63.4908 s
N=1000: 99.3897 s
```



Task B.3

Coarse-grained: cProfile (function-level)

```
import cProfile, pstats, sys, os
if os.path.isfile(os.path.join(os.getcwd(), "asm2", "conway_original.py")):
    sys.path.insert(0, os.path.join(os.getcwd(), "asm2"))
else:
    sys.path.insert(0, os.getcwd())
import conway_original

def benchmark():
    N, iters = 150, 50
    grid = conway_original.randomGrid(N)
    for _ in range(iters):
        grid = conway_original.step(grid, N)
pr = cProfile.Profile()
pr.enable()
benchmark()
pr.disable()
pstats.Stats(pr).sort_stats("cumulative").print_stats(15)
```

```
=== cProfile (coarse): top 15 by cumulative time (conway_original.py) ===

         158 function calls in 1.479 seconds

   Ordered by: cumulative time
```

```
   List reduced from 32 to 15 due to restriction <15>

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        2    0.000    0.000    1.479    0.740 /opt/miniconda3/envs/profiling/lib/python3.11/site-
packages/IPython/core/interactiveshell.py:3665(run_code)
        2    0.000    0.000    1.479    0.740 {built-in method builtins.exec}
        1    0.000    0.000    1.479    1.479
/var/folders/1w/8txt8lgn7ts1bxz2rxxd4y9c0000gn/T/ipykernel_41668/3108166553.py:16(benchmark)
       50    1.477    0.030    1.478    0.030 /Users/klaus/Documents/KTH/DD2358/dd2358-work/asm2/conway_original.py:59(step)
        1    0.001    0.001    0.001    0.001 /Users/klaus/Documents/KTH/DD2358/dd2358-
work/asm2/conway_original.py:19(randomGrid)
       50    0.001    0.000    0.001    0.000 {method 'copy' of 'numpy.ndarray' objects}
        2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/codeop.py:120(__call__)
        1    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-
packages/numpy/_core/fromnumeric.py:3287(prod)
        2    0.000    0.000    0.000    0.000 {built-in method builtins.compile}
        1    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-
packages/numpy/_core/fromnumeric.py:66(_wrapreduction)
        2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/contextlib.py:299(helper)
        2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/contextlib.py:104(__init__)
        1    0.000    0.000    0.000    0.000 {method 'reduce' of 'numpy.ufunc' objects}
        1    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-
packages/numpy/_core/getlimits.py:178(__new__)
        2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-
packages/traitlets/traitlets.py:676(__get__)









<pstats.Stats at 0x1228ead50>
```

```
=== cProfile (coarse): top 15 by cumulative time (conway_original.py) ===

         158 function calls in 1.479 seconds

   Ordered by: cumulative time
   List reduced from 32 to 15 due to restriction <15>

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        2    0.000    0.000    1.479    0.740 /opt/miniconda3/envs/profiling/lib/python3.11/site-packages/IPython/core/interactiveshell.py:3665(run_code
        2    0.000    0.000    1.479    0.740 {built-in method builtins.exec}
        1    0.000    0.000    1.479    1.479 /var/folders/1w/8txt8lgn7ts1bxz2rxxd4y9c0000gn/T/ipykernel_41668/3108166553.py:16(benchmark)
       50    1.477    0.030    1.478    0.030 /Users/klaus/Documents/KTH/DD2358/dd2358-work/asm2/conway_original.py:59(step)
        1    0.001    0.001    0.001    0.001 /Users/klaus/Documents/KTH/DD2358/dd2358-work/asm2/conway_original.py:19(randomGrid)
       50    0.001    0.000    0.001    0.000 {method 'copy' of 'numpy.ndarray' objects}
        2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/codeop.py:120(__call__)
        1    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-packages/numpy/_core/fromnumeric.py:3287(prod)
        2    0.000    0.000    0.000    0.000 {built-in method builtins.compile}
        1    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-packages/numpy/_core/fromnumeric.py:66(_wrapreduction)
        2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/contextlib.py:299(helper)
        2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/contextlib.py:104(__init__)
        1    0.000    0.000    0.000    0.000 {method 'reduce' of 'numpy.ufunc' objects}
        1    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-packages/numpy/_core/getlimits.py:178(__new__)
        2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-packages/traitlets/traitlets.py:676(__get__)
```

Fine-grained: line_profiler

```
import sys, os
if os.path.isfile(os.path.join(os.getcwd(), "asm2", "conway_original.py")):
    sys.path.insert(0, os.path.join(os.getcwd(), "asm2"))
else:
    sys.path.insert(0, os.getcwd())
import conway_original

%load_ext line_profiler
%lprun -f conway_original.step conway_original.step(conway_original.randomGrid(80), 80)
```

```
The line_profiler extension is already loaded. To reload it, use:
  %reload_ext line_profiler


Timer unit: 1e-09 s
```

```
Total time: 0.026689 s
File: /Users/klaus/Documents/KTH/DD2358/dd2358-work/asm2/conway_original.py
Function: step at line 59

Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
    59                                           def step(grid, N):
    60                                               """One simulation step (no visualization). For profiling; same logic as
inside update()."""
    61         1     586000.0 586000.0      2.2      newGrid = grid.copy()
    62        81      15000.0    185.2      0.1      for i in range(N):
    63      6480     492000.0     75.9      1.8          for j in range(N):
    64     12800    1482000.0    115.8      5.6              total = int(
    65      6400    1045000.0    163.3      3.9                  (
    66     51200    7091000.0    138.5     26.6                      grid[i, (j - 1) % N]
    67      6400    1498000.0    234.1      5.6                      + grid[i, (j + 1) % N]
    68      6400    1798000.0    280.9      6.7                      + grid[(i - 1) % N, j]
    69      6400    1628000.0    254.4      6.1                      + grid[(i + 1) % N, j]
    70      6400    2106000.0    329.1      7.9                      + grid[(i - 1) % N, (j - 1) % N]
    71      6400    1642000.0    256.6      6.2                      + grid[(i - 1) % N, (j + 1) % N]
    72      6400    1830000.0    285.9      6.9                      + grid[(i + 1) % N, (j - 1) % N]
    73      6400    1751000.0    273.6      6.6                      + grid[(i + 1) % N, (j + 1) % N]
    74                                                               )
    75      6400     621000.0     97.0      2.3                      / 255
    76                                                           )
    77      6400    1805000.0    282.0      6.8              if grid[i, j] == ON:
    78      1315     201000.0    152.9      0.8                  if (total < 2) or (total > 3):
    79       697     151000.0    216.6      0.6                      newGrid[i, j] = OFF
    80                                                           else:
    81      5085     595000.0    117.0      2.2                  if total == 3:
    82       758     352000.0    464.4      1.3                      newGrid[i, j] = ON
    83         1          0.0      0.0      0.0      return newGrid
```

```
Timer unit: 1e-09 s

Total time: 0.026689 s
File: /Users/klaus/Documents/KTH/DD2358/dd2358-work/asm2/conway_original.py
Function: step at line 59

Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
    59                                           def step(grid, N):
    60                                               """One simulation step (no visualization). For profiling; same logic as inside update()."""
    61         1     586000.0 586000.0      2.2      newGrid = grid.copy()
    62        81      15000.0    185.2      0.1      for i in range(N):
    63      6480     492000.0     75.9      1.8          for j in range(N):
    64     12800    1482000.0    115.8      5.6              total = int(
    65      6400    1045000.0    163.3      3.9                  (
    66     51200    7091000.0    138.5     26.6                      grid[i, (j - 1) % N]
    67      6400    1498000.0    234.1      5.6                      + grid[i, (j + 1) % N]
    68      6400    1798000.0    280.9      6.7                      + grid[(i - 1) % N, j]
    69      6400    1628000.0    254.4      6.1                      + grid[(i + 1) % N, j]
    70      6400    2106000.0    329.1      7.9                      + grid[(i - 1) % N, (j - 1) % N]
    71      6400    1642000.0    256.6      6.2                      + grid[(i - 1) % N, (j + 1) % N]
    72      6400    1830000.0    285.9      6.9                      + grid[(i + 1) % N, (j - 1) % N]
    73      6400    1751000.0    273.6      6.6                      + grid[(i + 1) % N, (j + 1) % N]
    74                                                               )
    75      6400     621000.0     97.0      2.3                      / 255
...
    79       697     151000.0    216.6      0.6                      newGrid[i, j] = OFF
    80                                                           else:
    81      5085     595000.0    117.0      2.2                  if total == 3:
    82       758     352000.0    464.4      1.3                      newGrid[i, j] = ON
    83         1          0.0      0.0      0.0      return newGrid
```

Task B.4

Optimized code (conway.py)

```python
"""
conway.py

A simple Python/matplotlib implementation of Conway's Game of Life.

Author: Mahesh Venkitachalam
"""

import sys, argparse
```

```python
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation

ON = 255
OFF = 0
vals = [ON, OFF]


def randomGrid(N):
    """returns a grid of NxN random values"""
    return np.random.choice(vals, N * N, p=[0.2, 0.8]).reshape(N, N)


def addGlider(i, j, grid):
    """adds a glider with top left cell at (i, j)"""
    glider = np.array([[0, 0, 255], [255, 0, 255], [0, 255, 255]])
    grid[i : i + 3, j : j + 3] = glider


def addGosperGliderGun(i, j, grid):
    """adds a Gosper Glider Gun with top left cell at (i, j)"""
    gun = np.zeros(11 * 38).reshape(11, 38)

    gun[5][1] = gun[5][2] = 255
    gun[6][1] = gun[6][2] = 255

    gun[3][13] = gun[3][14] = 255
    gun[4][12] = gun[4][16] = 255
    gun[5][11] = gun[5][17] = 255
    gun[6][11] = gun[6][15] = gun[6][17] = gun[6][18] = 255
    gun[7][11] = gun[7][17] = 255
    gun[8][12] = gun[8][16] = 255
    gun[9][13] = gun[9][14] = 255

    gun[1][25] = 255
    gun[2][23] = gun[2][25] = 255
    gun[3][21] = gun[3][22] = 255
    gun[4][21] = gun[4][22] = 255
    gun[5][21] = gun[5][22] = 255
    gun[6][23] = gun[6][25] = 255
    gun[7][25] = 255

    gun[3][35] = gun[3][36] = 255
    gun[4][35] = gun[4][36] = 255

    grid[i : i + 11, j : j + 38] = gun


def step(grid, N):
    # Neighbor count: sum 8 rolled views (toroidal wrap)
    rU = np.roll(grid, -1, axis=0)
    rD = np.roll(grid, 1, axis=0)
    rL = np.roll(grid, -1, axis=1)
    rR = np.roll(grid, 1, axis=1)
    rUL = np.roll(rU, -1, axis=1)
    rUR = np.roll(rU, 1, axis=1)
    rDL = np.roll(rD, -1, axis=1)
    rDR = np.roll(rD, 1, axis=1)
    n = (
        (rU + rD + rL + rR + rUL + rUR + rDL + rDR) / 255
    ).astype(np.int_)
    # Conway rules: birth (dead + 3 neighbors) or survive (live + 2 or 3)
    live = grid == ON
    birth = ~live & (n == 3)
    survive = live & ((n == 2) | (n == 3))
    return np.where(birth | survive, ON, OFF).astype(grid.dtype)


def update(frameNum, img, grid, N):
    """Animation callback: one step using optimized step()."""
    newGrid = step(grid, N)
```

```python
    img.set_data(newGrid)
    grid[:] = newGrid[:]
    return (img,)


# main() function
def main():
    # Command line args are in sys.argv[1], sys.argv[2] ..
    # sys.argv[0] is the script name itself and can be ignored
    # parse arguments
    parser = argparse.ArgumentParser(
        description="Runs Conway's Game of Life simulation."
    )
    # add arguments
    parser.add_argument("--grid-size", dest="N", required=False)
    parser.add_argument("--mov-file", dest="movfile", required=False)
    parser.add_argument("--interval", dest="interval", required=False)
    parser.add_argument("--glider", action="store_true", required=False)
    parser.add_argument("--gosper", action="store_true", required=False)
    args = parser.parse_args()

    # set grid size
    N = 100
    if args.N and int(args.N) > 8:
        N = int(args.N)

    # set animation update interval
    updateInterval = 50
    if args.interval:
        updateInterval = int(args.interval)

    # declare grid
    grid = np.array([])
    # check if "glider" demo flag is specified
    if args.glider:
        grid = np.zeros(N * N).reshape(N, N)
        addGlider(1, 1, grid)
    elif args.gosper:
        grid = np.zeros(N * N).reshape(N, N)
        addGosperGliderGun(10, 10, grid)
    else:
        # populate grid with random on/off - more off than on
        grid = randomGrid(N)

    # set up animation
    fig, ax = plt.subplots()
    img = ax.imshow(grid, interpolation="nearest")
    ani = animation.FuncAnimation(
        fig,
        update,
        fargs=(
            img,
            grid,
            N,
        ),
        frames=10,
        interval=updateInterval,
        save_count=50,
    )

    # # of frames?
    # set output file
    if args.movfile:
        ani.save(args.movfile, fps=30, extra_args=["-vcodec", "libx264"])

    plt.show()


# call main
if __name__ == "__main__":
```

```
    main()
```

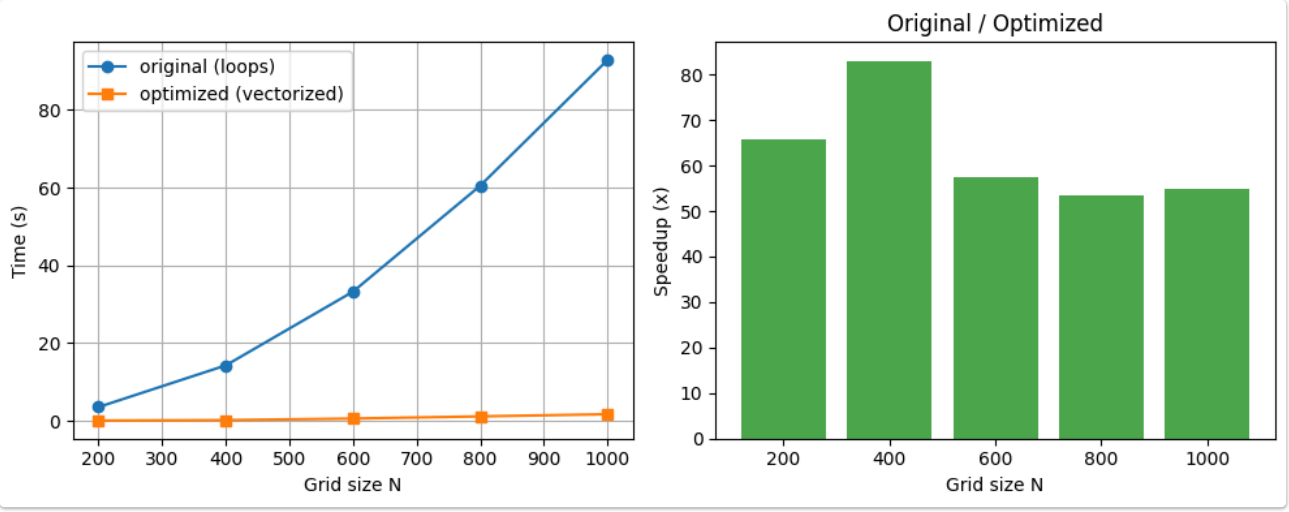
Performance comparison: conway_original.py vs conway.py (optimized)

```
import sys, os, time
import matplotlib.pyplot as plt

if os.path.isfile(os.path.join(os.getcwd(), "asm2", "conway_original.py")):
    sys.path.insert(0, os.path.join(os.getcwd(), "asm2"))
else:
    sys.path.insert(0, os.getcwd())
import conway_original
import conway

num_iterations = 100
grid_sizes = [200, 400, 600, 800, 1000]
times_orig = []
times_opt = []
for N in grid_sizes:
    grid = conway.randomGrid(N)
    t0 = time.perf_counter()
    g = grid.copy()
    for _ in range(num_iterations):
        g = conway_original.step(g, N)
    times_orig.append(time.perf_counter() - t0)
    t0 = time.perf_counter()
    g = grid.copy()
    for _ in range(num_iterations):
        g = conway.step(g, N)
    times_opt.append(time.perf_counter() - t0)
speedups = [times_orig[i] / times_opt[i] for i in range(len(grid_sizes))]
print("N    | original (s) | optimized (s) | speedup")
print("-" * 50)
for i, N in enumerate(grid_sizes):
    print(f"{N:4} | {times_orig[i]:12.4f} | {times_opt[i]:13.4f} | {speedups[i]:.1f}x")
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))
ax1.plot(grid_sizes, times_orig, "o-", label="original (loops)")
ax1.plot(grid_sizes, times_opt, "s-", label="optimized (vectorized)")
ax1.set_xlabel("Grid size N")
ax1.set_ylabel("Time (s)")
ax1.legend()
ax1.grid(True)
ax2.bar(range(len(grid_sizes)), speedups, color="green", alpha=0.7)
ax2.set_xticks(range(len(grid_sizes)))
ax2.set_xticklabels(grid_sizes)
ax2.set_xlabel("Grid size N")
ax2.set_ylabel("Speedup (x)")
ax2.set_title("Original / Optimized")
plt.tight_layout()
plt.show()
```

```
N    | original (s) | optimized (s) | speedup
--------------------------------------------------
 200 |       3.5374 |        0.0539 | 65.6x
 400 |      14.2603 |        0.1719 | 83.0x
 600 |      33.2428 |        0.5774 | 57.6x
 800 |      60.5117 |        1.1346 | 53.3x
1000 |      92.8445 |        1.6878 | 55.0x
```




```
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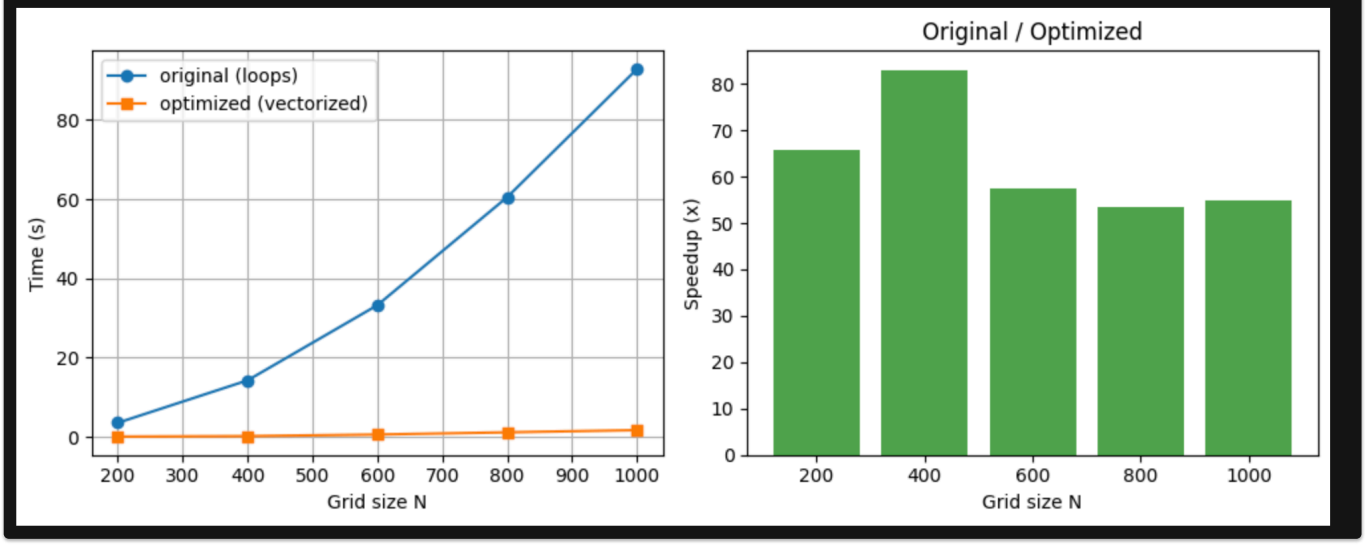

New profiling results

```python
import cProfile, pstats, sys, os
if os.path.isfile(os.path.join(os.getcwd(), "asm2", "conway.py")):
    sys.path.insert(0, os.path.join(os.getcwd(), "asm2"))
else:
    sys.path.insert(0, os.getcwd())
import conway

def run():
    grid = conway.randomGrid(150)
    for _ in range(50):
        grid = conway.step(grid, 150)
pr = cProfile.Profile()
pr.enable()
run()
pr.disable()
pstats.Stats(pr).sort_stats("cumulative").print_stats(12)
```

```
         158 function calls in 1.594 seconds

   Ordered by: cumulative time
   List reduced from 32 to 12 due to restriction <12>

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        2    0.000    0.000    1.593    0.797 /opt/miniconda3/envs/profiling/lib/python3.11/site-
packages/IPython/core/interactiveshell.py:3665(run_code)
```

```
        2    0.000    0.000    1.593    0.797 {built-in method builtins.exec}
        1    0.001    0.001    1.593    1.593
/var/folders/1w/8txt8lgn7ts1bxz2rxxd4y9c0000gn/T/ipykernel_41668/255385970.py:8(run)
       50    1.565    0.031    1.588    0.032 /Users/klaus/Documents/KTH/DD2358/dd2358-work/asm2/conway.py:59(step)
       50    0.023    0.000    0.023    0.000 {method 'copy' of 'numpy.ndarray' objects}
        1    0.003    0.003    0.003    0.003 /Users/klaus/Documents/KTH/DD2358/dd2358-work/asm2/conway.py:19(randomGrid)
        1    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-
packages/numpy/_core/fromnumeric.py:3287(prod)
        2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/codeop.py:120(__call__)
        1    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-
packages/numpy/_core/fromnumeric.py:66(_wrapreduction)
        2    0.000    0.000    0.000    0.000 {built-in method builtins.compile}
        1    0.000    0.000    0.000    0.000 {method 'reduce' of 'numpy.ufunc' objects}
        2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/contextlib.py:132(__enter__)





<pstats.Stats at 0x123045b10>
```

```
      158 function calls in 1.594 seconds

Ordered by: cumulative time
List reduced from 32 to 12 due to restriction <12>

ncalls  tottime  percall  cumtime  percall filename:lineno(function)
     2    0.000    0.000    1.593    0.797 /opt/miniconda3/envs/profiling/lib/python3.11/site-packages/IPython/core/interactiveshell.py:3665(run_code)
     2    0.000    0.000    1.593    0.797 {built-in method builtins.exec}
     1    0.001    0.001    1.593    1.593 /var/folders/1w/8txt8lgn7ts1bxz2rxxd4y9c0000gn/T/ipykernel_41668/255385970.py:8(run)
    50    1.565    0.031    1.588    0.032 /Users/klaus/Documents/KTH/DD2358/dd2358-work/asm2/conway.py:59(step)
    50    0.023    0.000    0.023    0.000 {method 'copy' of 'numpy.ndarray' objects}
     1    0.003    0.003    0.003    0.003 /Users/klaus/Documents/KTH/DD2358/dd2358-work/asm2/conway.py:19(randomGrid)
     1    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/site-packages/numpy/_core/fromnumeric.py:3287(prod)
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     2    0.000    0.000    0.000    0.000 /opt/miniconda3/envs/profiling/lib/python3.11/contextlib.py:132(__enter__)
```

AI Usage

Basic Part

AI is not used in the basic part.

Bonus Part

Composer 1.5

Task B.2

Suggested code to plot the graph

Task B.3

Brainstorming for what coarse and fine-grained profiler I can use

Task B.4

Brainstorming: Ways to further optimization, Suggested code to plot the graph

All edited and modified by us.

Prompts:

Task B.2

"Based on the code I have writen, suggest the code to plot a graph"

Task B.3

"Suggest some coarse and fine-grained profiler with Python for free that I can use"

Task B.4

"Based on the code I provided, can it be further optimized?"
"Based on the code I provided, suggest the code to plot a graph to compare the result with the original and the optimized code."

Validation:

Task B.2

Modify some format

Task B.3

Choose cProfile and line_profiler which we have used before

Task B.4

Plot and experiment that the suggestion of boolean indexing for the live or dead rools further optimized the code